CLASS 8 — MATHS · MEDIUM LEVEL CHAPTER TEST SET 2

Chapter 11: **Exploring Some Geometric Themes** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A polyhedron has 32 faces and 90 edges. Its number of vertices is: 1

(a) 120 **(b)** 58 **(c)** 124 **(d)** 60

2. A cube of side 6 cm is painted all over and cut into 1 cm cubes. The number of cubes with no face painted is: 1

(a) 96 **(b)** 64 **(c)** 208 **(d)** 16

3. A Koch curve starts with a line of length 81 cm. The total length after stage 3 (in cm) is: 1

(a) 192 **(b)** 324 **(c)** 243 **(d)** 34.17

4. The total number of triangles removed from the Sierpinski triangle up to stage 4 is: 1

(a) 81 **(b)** 80 **(c)** 27 **(d)** 40

5. A prism has 27 edges. The number of its faces is: 1

(a) 10 **(b)** 11 **(c)** 9 **(d)** 12

6. A pyramid has 11 vertices. The number of its edges is: 1

(a) 12 **(b)** 10 **(c)** 22 **(d)** 20

7. All 14 faces of a polyhedron are triangles. The number of its edges is: 1

(a) 14 **(b)** 42 **(c)** 21 **(d)** 28

8. A polyhedron has 14 faces and 36 edges. Its number of vertices is: 1

(a) 48 **(b)** 24 **(c)** 52 **(d)** 22

9. A cube of side 8 cm is painted all over and cut into 1 cm cubes. The number of cubes with no face painted is: 1

(a) 504 **(b)** 343 **(c)** 216 **(d)** 36

10. A Koch curve starts with a line of length 189 cm. The total length after stage 3 (in cm) is: 1

(a) 448 **(b)** 79.73 **(c)** 567 **(d)** 756

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. A prism has 39 edges. Find its number of faces and vertices. 2

12. A Koch curve starts with a line of length 72 cm. Find the number of segments and the total length at stage 2. 2

13. A cuboid is 18 cm long, 11 cm wide and 7 cm high. Find the sum of lengths of all its edges and the number of edges. 2

14. A prism has 45 edges. Find its number of faces and vertices. 2

SECTION — C (Short Answer Type) 4 × 3 = 12

Answer in about 40–60 words each.

15. A cube of side 6 cm is painted all over and cut into 1 cm cubes. Find the number of cubes with 3
(i) exactly three painted faces
(ii) no painted face
(iii) at least one painted face.

16. All 22 faces of a polyhedron are triangles. Find the number of edges and vertices. 3

17. A Sierpinski triangle starts with a triangle of area 1280 cm². Find the shaded area and the number of shaded triangles at stage 4. 3

OR

A cube of side 7 cm is painted all over and cut into 1 cm cubes. Find the number of cubes with 3
(i) exactly three painted faces
(ii) no painted face
(iii) at least one painted face.

18. All 26 faces of a polyhedron are triangles. Find the number of edges and vertices. 3

OR

A Sierpinski triangle starts with a triangle of area 576 cm². Find the shaded area and the number of shaded triangles at stage 2. 3

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 100–150 words each.

19. A prism and a pyramid both have a heptagonal base. Find the difference between the number of 5
(i) faces
(ii) vertices
(iii) edges of the prism and the pyramid.

OR

A Koch snowflake is built from an equilateral triangle of side 27 cm. Find its perimeter at stage 0, stage 1 and stage 2. 5

20. A prism and a pyramid both have a octagonal base. Find the difference between the number of 5
(i) faces
(ii) vertices
(iii) edges of the prism and the pyramid.

OR

A Koch snowflake is built from an equilateral triangle of side 36 cm. Find its perimeter at stage 0, stage 1 and stage 2. 5

END OF QUESTION PAPER Total: 40 Marks

MASTERMIND